

Harshavardhan Reddy Kaditham

+1 930-904-2942

kharsha9990@gmail.com

LinkedIn

GitHub

Education

Indiana University Bloomington

Master's in Data Science (GPA: 3.57/4.00)

Bloomington, IN

Aug 2024 – May 2026

Sri Venkateshwara College of Engineering

B.E. in Computer Science (GPA: 8.55/10)

Bangalore, India

Dec 2020 – May 2024

Technical Skills

Programming: Python, R, SQL, C/C++

AI/ML: PyTorch, Scikit-learn, HuggingFace Transformers, LangChain, LangGraph, BERT, GPT, LLaMA, RAG, LoRA/QLoRA, RLHF, Embeddings, Prompt Engineering

Data Science: Feature Engineering, Predictive Modeling, EDA, Model Evaluation, Statistical Analysis

Visualization: Tableau, Power BI, Matplotlib, Seaborn, Plotly, ggplot2

Databases & Cloud: MySQL, PostgreSQL, MongoDB, Neo4j, Spark, PySpark, AWS, GCP, FAISS, ChromaDB

Professional Experience

Zummit Infolabs

Sep 2023 – Dec 2023

Data Science Intern

Bangalore, India

- Built **NLP and Computer Vision pipelines** for recruitment automation and car scratch detection, reducing manual screening time by **30%** across production workflows.
- Designed **deep learning prototypes** for emotion sincerity detection and autopilot systems using **Python and OpenCV**, improving model inference speed by **20%**.
- Accelerated model iteration across **3+ AI use cases** through feature engineering, **SMOTE** balancing, and evaluation optimization, improving deployment readiness.

TechnoHacks

Aug 2023 – Sep 2023

Data Science Intern

Bangalore, India

- Processed **10K+ records** for churn prediction, housing price estimation, and sentiment analysis using EDA, preprocessing, and predictive modeling in Python.
- Reduced model error rates by **15%** by tuning regression, classification, and NLP pipelines with optimized feature selection and cross-validation workflows.
- Built Python-based dashboards to track model performance metrics, improving reporting efficiency across **3 data science projects**.

Projects

AI Resume Screener with LLMs (Python, LangChain, RAG, FAISS, OpenAI API, HuggingFace, Streamlit, MySQL)

- Engineered an **LLM-powered recruitment pipeline** using **RAG and LangChain** to match and rank **500+ resumes** against job descriptions, generating explainable candidate fit scores via **semantic similarity scoring**.
- Built a **FAISS vector store** to embed and retrieve resume chunks, reducing irrelevant candidate retrieval by **~40%** compared to keyword-based screening baselines.
- Designed a **multi-stage evaluation layer** with structured LLM output parsing, automated gap analysis, and recruiter-ready reports served via an **interactive Streamlit dashboard** backed by MySQL.

Fine-Tuned LLM for Medical Text Summarization (Python, PyTorch, HuggingFace, LLaMA, LoRA, QLoRA, Gradio)

- Fine-tuned a **LLaMA-based model** on **10K+ clinical notes** using **LoRA and QLoRA**, reducing trainable parameters by **90%** while achieving a **ROUGE-L score of 0.43** on medical summarization benchmarks.
- Built an **end-to-end training pipeline** with HuggingFace Transformers and PyTorch, incorporating custom tokenization, data loaders, and evaluation loops for domain-specific medical text.
- Deployed an **interactive Gradio interface** for real-time clinical note summarization, enabling structured discharge summary generation from unstructured patient records.

AI Data Analyst Agent (Python, LangChain, LangGraph, OpenAI API, Pandas, Plotly, Streamlit, SQLite)

- Built an **agentic data analysis pipeline** using **LangGraph** where a reasoning agent autonomously generates and executes **Python and SQL code** in response to natural language queries over uploaded CSV and database files.
- Integrated **tool-calling** with Pandas and Plotly to enable dynamic chart generation, statistical summaries, and multi-step data transformations, reducing manual analysis time by **~60%**.
- Deployed a **Streamlit interface** supporting multi-turn conversations over live datasets, with structured output parsing and **execution sandboxing** for safe and reliable code runs.

Certifications

- IBM Generative AI Engineering with LLMs
- IBM Data Science Professional Certificate
- Python for Data Science and Machine Learning Bootcamp

Awards & Achievements

- TechEdon 2.0 Hackathon (1st Place):** Architected FinIt, a finance management application for smart budgeting and expense tracking.